COM203 - Computer Architecture



Lab # 4

Mars





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Outline

- PRINTING A DOUBLE
- ADDING TWO INTEGERS



PRINTING A DOUBLE

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Code:

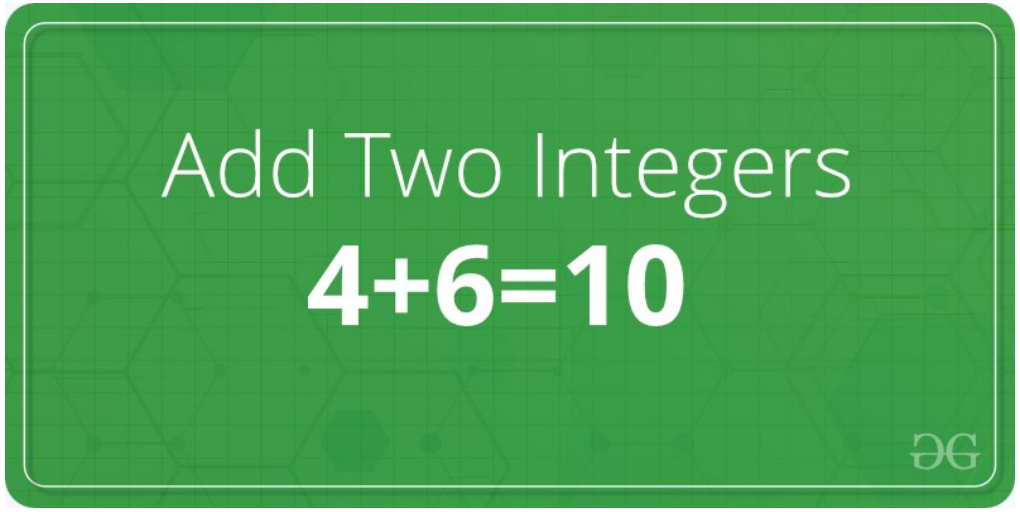
```
.data
   myDouble:   .double 3.141
   ZeroDouble: .double 0.0

.text
   ldc1 $f2, myDouble
   ldc1 $f0, ZeroDouble



   li $v0, 3
   add.d $f12, $f2, $f0          # add to f0 to f2 and store in f12 (put value in argument)
   syscall
      li $v0, 10                 #Exit command
syscall
```

ADDING TWO INTEGERS



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Code:

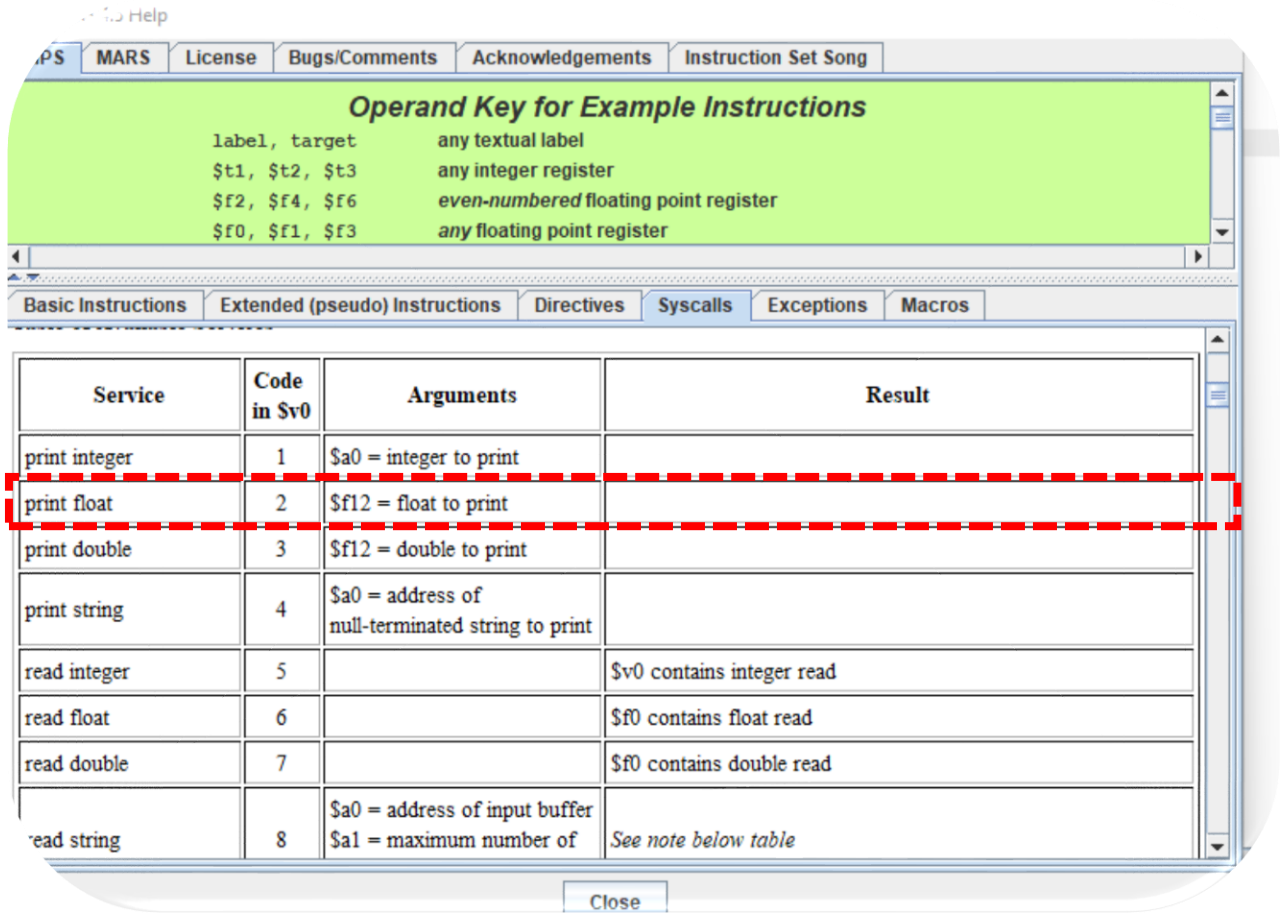
```
.data
    number1:  .word 8
    number2:  .word 18

.text
  lw $t0, number1 ($zero)
  lw $t1, number2 ($zero)

  add $t2, $t0, $t1            #t2= t0+t1

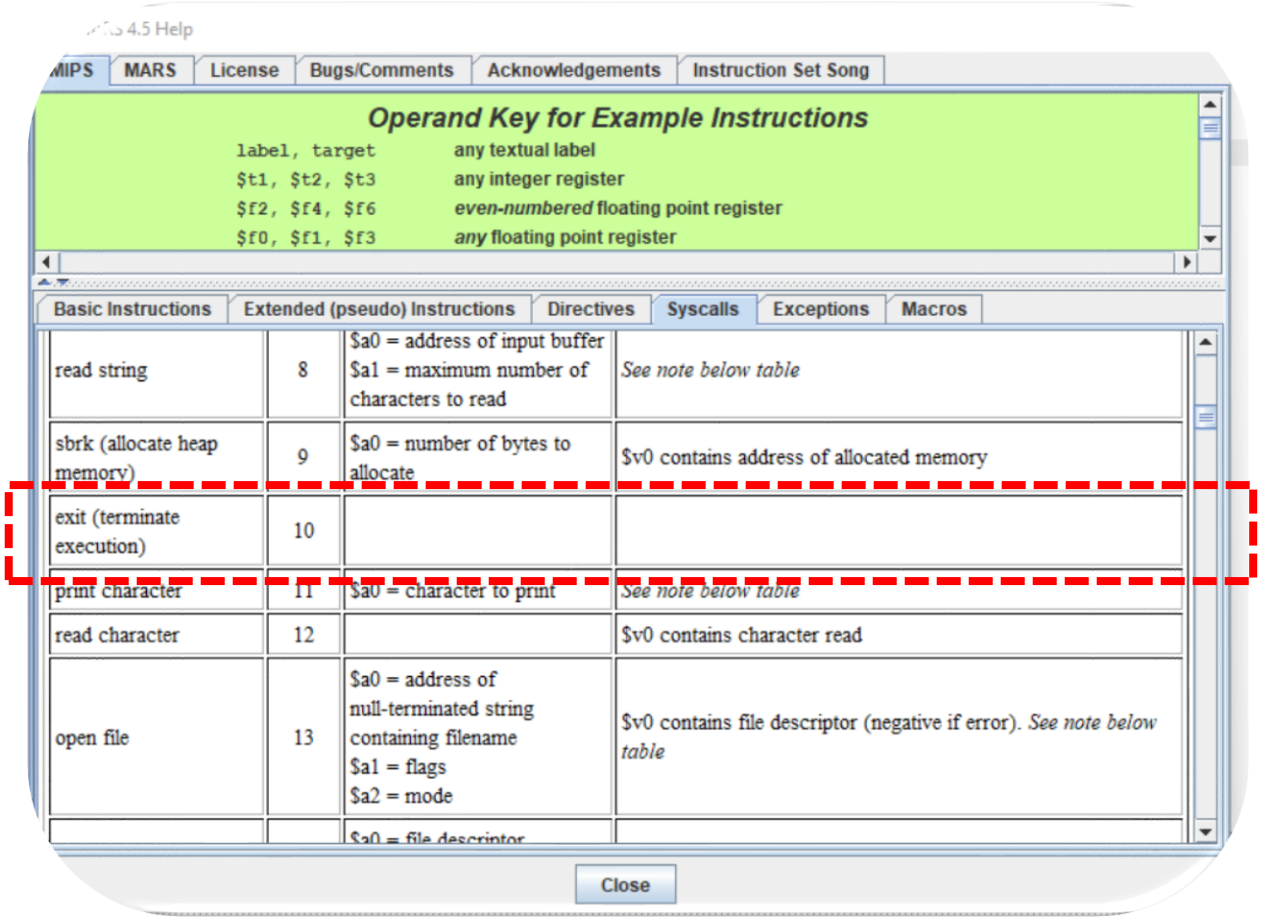
  li $v0, 1                       #print
  add $a0, $zero, $t2
  syscall
      li $v0, 10                  #Exit command
syscall
```

Print FLOAT



Service	Code in $v0	Arguments	Result
print integer	1	$a0 = integer to print	
print float	2	$f12 = float to print	
print double	3	$f12 = double to print	
print string	4	$a0 = address of null-terminated string to print	
read integer	5		$v0 contains integer read
read float	6		$f0 contains float read
read double	7		$f0 contains double read
ead string	8	$a0 = address of input buffer $a1 = maximum number of	*See note below table*

EXIT


MIPS
MARS
License
Bugs/Comments
Acknowledgements
Instruction Set Song
Operand Key for Example Instructions
label, target any textual label
$t1, $t2, $t3 any integer register
$f2, $f4, $f6 even-numbered floating point register
$f0, $f1, $f3 any floating point register
Basic Instructions
Extended (pseudo) Instructions
Directives
Syscalls
Exceptions
Macros
read string 8 $a0 = address of input buffer $a1 = maximum number of characters to read See note below table
sbrk (allocate heap memory) 9 $a0 = number of bytes to allocate $v0 contains address of allocated memory
exit (terminate execution) 10
print character 11 $a0 = character to print See note below table
read character 12 $v0 contains character read
open file 13 $a0 = address of null-terminated string containing filename $a1 = flags $a2 = mode $v0 contains file descriptor (negative if error). See note below table
Close

End of LAB SESSION! ☺

- **Next week:** SUBTRACTION TWO INTEGER

- **Questions?**

